The Explanation Capacity Audit: Standard ML Explanation Practice Routinely Exceeds the Limits of Explanation

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Abstract

The Explanation Capacity Theorem proves that for any system with symmetry group G acting on a feature space V, the maximum number of independent stable explanatory facts is $C = \dim(V^G)$—the dimension of the G-invariant subspace. Any claim beyond C has mean squared error equal to the full norm of the target (the over-explanation penalty). We audit 149 datasets across 53 scientific domains and find that 75% exceed their explanation capacity under standard ML explanation practice: the feature rankings reported in typical SHAP analyses contain more pairwise claims than the correlation structure can support. The directional prediction—within-group pairwise comparisons are less stable than between-group comparisons—is confirmed in 86% of testable datasets (27 significant confirmations, 1 significant reversal; 19 Bonferroni-corrected). The effect is robust across model classes (XGBoost, Random Forest, Ridge; 80% cross-model agreement) and explanation methods (TreeSHAP $\approx$ native importance). The single reversal (Dermatology) identifies a principled scope condition: the prediction holds for substitutable features competing for signal, not for complementary features co-occurring as disease indicators. These results establish that the gap between what ML models can predict and what can be stably explained about them is not a deficiency of current methods but a mathematical limit—one that 75% of standard practice exceeds.

1 Introduction

A machine learning model trained on gene expression data names TSPAN8 as the most important biomarker for distinguishing colon from kidney tumours. Retrain with a different random seed: the model now names CEACAM5, an immune-evasion marker sharing zero Gene Ontology terms with TSPAN8. Both models achieve indistinguishable accuracy [**?**].

This instability is not a software bug. The Explanation Capacity Theorem [**?**] proves that for any system with correlated features, there is a hard upper bound—the *explanation capacity* $C = \dim(V^G)$—on the number of independent feature-level claims that any method can stably support. Claims beyond C have mean squared error equal to the full norm of the target: the best stable approximation to beyond-capacity content is the zero vector [**?**].

How often does published practice exceed this bound? This paper answers the question empirically. We audit 149 datasets from 53 scientific domains—genomics, neuroscience, particle physics, manufacturing, veterinary medicine, and 48 others—and find that 75% exceed their explanation capacity under standard SHAP analysis with stochastic regularisation.

2 Methods

2.1 The Explanation Capacity

For a feature space V with P features falling into g correlation groups under the symmetry group G (identified by SAGE: hierarchical clustering on |Spearman ρ| at threshold $\rho^* = 0.70$), the explanation capacity is $C = g$. A standard SHAP analysis ranking all P features makes P positional claims. The dataset *exceeds capacity* if $P > C$.

2.2 Audit Protocol

For each of 149 datasets (from scikit-learn, OpenML, and PMLB):

1. Train $M = 50$ XGBoost models with stochastic regularisation (subsample = 0.8, colsample_bytree = 0.5, different random seeds)
2. Compute TreeSHAP importance (mean |SHAP|) for each model
3. Identify correlation groups via SAGE at $\rho^* = 0.70$
4. Compute pairwise flip rates: for each feature pair (i, j), the fraction of seeds where the minority ranking occurs
5. Classify each pair as *within-group* (features in the same SAGE group; predicted unstable) or *between-group* (different groups; predicted stable)
6. Test: within-group flip rate > between-group flip rate (Mann–Whitney U, one-sided, $\alpha = 0.005$)

Sensitivity analysis at $\rho^* \in \{0.50, 0.60, 0.70, 0.80, 0.90\}$. Model-class robustness tested with Random Forest and Ridge on 5 representative datasets. Method robustness tested with native feature importance vs. TreeSHAP.

3 Results

3.1 Exceedance

Of 149 datasets, 111 (74.5%) exceed their explanation capacity at $\rho^* = 0.70$. The 38 non-exceeding datasets are predominantly cases where features are approximately independent ($g \approx P$), confirming the theorem's scope condition: no Rashomon property $\Rightarrow$ no exceedance.

Exceedance is robust to threshold choice: 97% at $\rho^* = 0.50$, 94% at 0.60, 75% at 0.70, 70% at 0.80, 51% at 0.90.

3.2 Directional Prediction

Of 84 datasets with measurable within-group pairs, 72 (85.7%) show the predicted direction (within-group flip rate > between-group flip rate). 27 are significant at $p < 0.005$ (19 Bonferroni-corrected at $p < 4.4 \times 10^{-5}$). One dataset shows a significant reversal (Dermatology; see Section 3.6).

The 27:1 confirmation-to-reversal ratio at $p < 0.005$ is the headline result. Under the null hypothesis of no directional effect, the expected ratio is 1:1.

3.3 Effect Sizes

The strongest confirmations span diverse domains: Satellite Image (remote sensing, $d = 1.27$), HAR Activity (wearables, $d = 0.91$), Horse Colic (veterinary, $d = 0.99$), AP Colon Kidney (genomics, $d = 0.90$), MiniBooNE (particle physics, $d = 0.92$), Vehicle Silhouette (vehicle engineering, $d = 0.82$), MiceProtein (neuroscience, $d = 0.67$).

Median Cohen's d across all testable datasets: 0.40 (medium effect).

3.4 η Law Degradation

The within-group flip rate increases monotonically with within-group correlation strength on synthetic data with known groups ($r = 0.898$, $p = 0.006$). The bimodal gap (within − between) also increases with ρ ($r = 0.864$, $p = 0.012$). At $\rho = 0.99$ (near-exact exchangeability), the gap is +0.31; at $\rho = 0.50$ (threshold boundary), groups are undetectable.

This confirms the η law's degradation curve: the prediction is sharpest when features are exactly exchangeable and degrades continuously with approximate symmetry.

3.5 Robustness

Model class: XGBoost, Random Forest, and Ridge agree on the direction of the within/between flip rate difference in 4 of 5 test datasets (80%). The one disagreement (Diabetes) has Ridge showing a tiny reversal ($d = -0.33$) while XGBoost and RF both confirm.

Explanation method: TreeSHAP and native XGBoost feature importance agree on direction in all 3 tested datasets. Native importance shows stronger effects in some cases (Diabetes: $d = 1.81$ native vs. 0.94 TreeSHAP).

3.6 The Dermatology Reversal

The single significant reversal is Dermatology ($P = 34$, $g = 22$, within $= 0.039$, between $= 0.137$, $d = -0.80$, reversal $p = 3.6 \times 10^{-4}$). The reversal persists across all thresholds ($\rho^* = 0.50$ to 0.90), confirming it is structural.

The explanation: dermatology features are clinical indicators that *co-occur* (scaling + erythema indicate the same disease), not features that *compete* for signal (as correlated gene expression features do). The theorem's prediction requires substitutability: correlated features must be interchangeable with respect to the model. When correlated features are functionally complementary—used jointly by the model rather than selected among—within-group stability is *higher* than between-group stability.

This identifies a principled scope condition for the correlation-as-proxy-for-exchangeability approach: the SAGE grouping correctly identifies substitutable features (the common case in high-dimensional continuous data) but misidentifies complementary features (the uncommon case in clinical indicator data).

4 Discussion

The audit establishes three results.

First, **standard ML explanation practice routinely exceeds the explanation capacity**. 75% of 149 datasets—spanning 53 domains from genomics to particle physics—have more feature-level ranking claims than the correlation structure can stably support. This is not a critique of any

particular study; it is a structural property of how SHAP analyses are reported when features are correlated.

Second, **the capacity law's directional prediction is empirically confirmed across diverse domains**. Within-group comparisons are less stable than between-group comparisons in 86% of testable datasets, with 27 significant confirmations and 1 significant reversal (27:1 ratio). The effect is robust across model classes and explanation methods.

Third, **the framework correctly identifies its own scope**. The 38 non-exceeding datasets are cases where features are independent (no Rashomon property). The 1 reversal is a case where correlation indicates complementarity rather than substitutability. Both are predicted by the theory's preconditions.

Implications for practice. The audit does not claim that published studies with exceedance are "wrong." Group-level conclusions may be correct even when feature-level specificity is unsupported. A study reporting "TSPAN8 is the most important gene" is over-specified; a study reporting "the invasion pathway genes are the most important group" may be perfectly stable.

The recommendation is a *stability audit*: before reporting feature-level explanations, compute the SAGE groups, the capacity C, and report only the C group-level facts as definitive. Flag within-group rankings as provisional. This costs minutes and prevents the systematic publication of beyond-capacity claims as stable findings.

Limitations. The audit uses correlation as a proxy for exchangeability. The Dermatology reversal demonstrates that this proxy fails for complementary features. The audit covers only XGBoost with TreeSHAP as the primary method; robustness checks on RF and Ridge are limited to 5 datasets. The within-group flip rates (mean 0.22) are below the exact-exchangeability prediction of 0.50 because the audit datasets have moderate, not exact, correlation. The 149 datasets are a convenience sample from sklearn, OpenML, and PMLB, not a random sample of published ML studies.

Data and Code Availability

All code, results, and the pre-registration protocol are available at `https://github.com/DrakeCaraker/universal-explanation-impossibility`.